Assignment - 1

Code -

```
from email import message_from_string
from email.utils import parsedate_to_datetime

print("========== EMAIL HEADER ANALYZER ==========")

header = """From: sender@example.com
To: receiver@example.com
Subject: Urgent Payment Request
Date: Mon, 12 Oct 2026 10:30:00 +0530
Received: from mail.example.com (192.168.1.10)
"""

msg = message_from_string(header)

print("\n========== EMAIL HEADER ANALYSIS ==========")
print("Sender         :", msg["From"])
print("Receiver       :", msg["To"])
print("Subject        :", msg["Subject"])
print("Date           :", msg["Date"])

received = msg.get("Received", "")

if received:
    print("Received Info  :", received)

    import re
    ips = re.findall(r'\b(?:\d{1,3}\.){3}\d{1,3}\b', received)

    if ips:
        print("IP Address     :", ips[0])
        print("Note           : Check whether the IP is public or private.")
    else:
        print("IP Address     : Not found")
else:
    print("Received Info  : Not available")

print("\n========== INVESTIGATION REPORT ==========")
print("Email header analyzed successfully.")
print("Available header information extracted.")
print("Analysis completed.")
```

Output -

Output

```
========== EMAIL HEADER ANALYZER ==========

========== EMAIL HEADER ANALYSIS ==========
Sender          : sender@example.com
Receiver        : receiver@example.com
Subject         : Urgent Payment Request
Date            : Mon, 12 Oct 2026 10:30:00 +0530
Received Info   : from mail.example.com (192.168.1.10)
IP Address      : 192.168.1.10
Note            : Check whether the IP is public or private.

========== INVESTIGATION REPORT ==========
Email header analyzed successfully.
Available header information extracted.
Analysis completed.
```

Assignment - 3

Code -

```
from datetime import datetime

# Sample log entries
logs = [
    "2026-10-13 10:00:01 INFO User login successful",
    "2026-10-13 10:01:15 WARNING Multiple failed login attempts",
    "2026-10-13 10:01:20 ERROR Unauthorized access detected",
    "2026-10-13 10:02:10 INFO User logout successful",
    "2026-10-13 10:03:00 ERROR Database connection failed"
]

print("========== LOG CAPTURING ==========")

for log in logs:
    print(log)

print("\n========== EVENT CORRELATION ==========")

failed_attempts = 0
errors = 0

for log in logs:
    if "WARNING" in log:
        failed_attempts += 1

    if "ERROR" in log:
        errors += 1

print("Total log entries:", len(logs))
print("Warning events:", failed_attempts)
print("Error events:", errors)

if failed_attempts > 0 and errors > 0:
    print("\nAlert: Suspicious activity detected!")
    print("Warning events and errors found in logs.")
else:
    print("\nNo suspicious activity detected.")

print("\nLog analysis completed.")
```

Output -

Output

```
========== LOG CAPTURING ==========
2026-10-13 10:00:01 INFO User login successful
2026-10-13 10:01:15 WARNING Multiple failed login attempts
2026-10-13 10:01:20 ERROR Unauthorized access detected
2026-10-13 10:02:10 INFO User logout successful
2026-10-13 10:03:00 ERROR Database connection failed

========== EVENT CORRELATION ==========
Total log entries: 5
Warning events: 1
Error events: 2

Alert: Suspicious activity detected!
Warning events and errors found in logs.

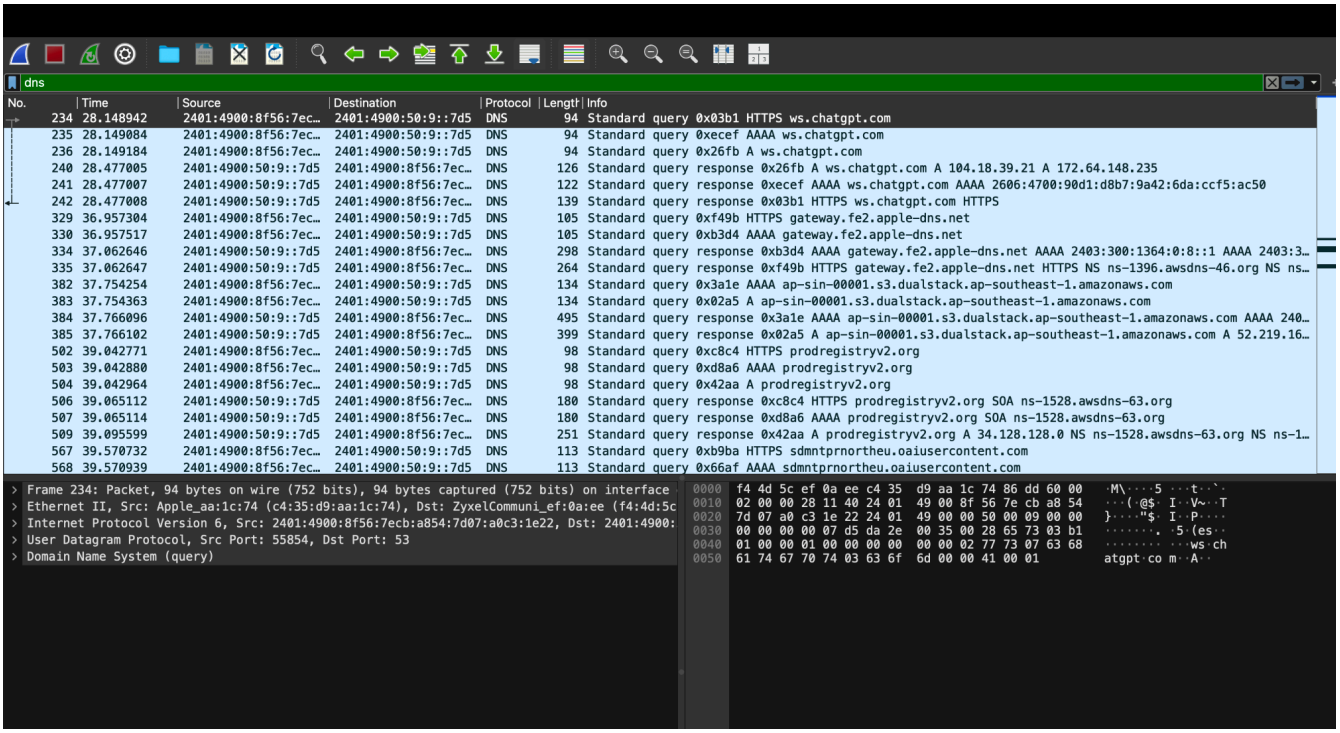
Log analysis completed.
```

Assignment - 4

Steps to Perform the Practical

1. **Download Wireshark:** Visit the official website, https://www.wireshark.org/download.html, and download Wireshark for your operating system. Install it by following the on-screen instructions.
2. **Configure Wireshark:** Open Wireshark after installation. If a permission warning appears on macOS, click the provided link to install ChmodBPF and enable packet-capture permissions.
3. **Select Network Interface:** Identify the active network interface, such as Wi-Fi (en0), from the available interfaces. Double-click it to start capturing network packets.
4. **Generate Network Traffic:** Keep Wireshark running and open a web browser. Visit a website to generate network traffic. Observe the packets appearing in the Wireshark window.
5. **Stop Packet Capture:** Allow the capture to run for a few seconds, then click the red square button in the toolbar to stop capturing packets.
6. **Apply Display Filters:** Enter `dns` in the display filter bar and press Enter to display DNS packets. You can also use the `tcp` filter to view TCP packets or `tls` to view TLS traffic.
7. **Analyze Captured Packets:** Examine the packet list, including packet number, time, source address, destination address, and protocol. Select a packet to view its detailed information.
8. **Save the Capture File:** Click **File** → **Save As**, enter a suitable filename, such as `Practical6_Capture.pcapng`, select a folder, and click Save.
9. **Record the Output:** Take screenshots showing the captured packets and applied display filter. Insert the screenshots into your practical journal as evidence of successful execution.

Output -


dns
No. Time Source Destination Protocol Lengtl Info
234 28.148942 2401:4900:8f56:7ec… 2401:4900:50:9::7d5 DNS 94 Standard query 0x03b1 HTTPS ws.chatgpt.com
235 28.149084 2401:4900:8f56:7ec… 2401:4900:50:9::7d5 DNS 94 Standard query 0xecef AAAA ws.chatgpt.com
236 28.149184 2401:4900:8f56:7ec… 2401:4900:50:9::7d5 DNS 94 Standard query 0x26fb A ws.chatgpt.com
240 28.477005 2401:4900:50:9::7d5 2401:4900:8f56:7ec… DNS 126 Standard query response 0x26fb A ws.chatgpt.com A 104.18.39.21 A 172.64.148.235
241 28.477007 2401:4900:50:9::7d5 2401:4900:8f56:7ec… DNS 122 Standard query response 0xecef AAAA ws.chatgpt.com AAAA 2606:4700:90d1:d8b7:9a42:6da:ccf5:ac50
242 28.477008 2401:4900:50:9::7d5 2401:4900:8f56:7ec… DNS 139 Standard query response 0x03b1 HTTPS ws.chatgpt.com HTTPS
329 36.957304 2401:4900:8f56:7ec… 2401:4900:50:9::7d5 DNS 105 Standard query 0xf49b HTTPS gateway.fe2.apple-dns.net
330 36.957517 2401:4900:8f56:7ec… 2401:4900:50:9::7d5 DNS 105 Standard query 0xb3d4 AAAA gateway.fe2.apple-dns.net
334 37.062646 2401:4900:50:9::7d5 2401:4900:8f56:7ec… DNS 298 Standard query response 0xb3d4 AAAA gateway.fe2.apple-dns.net AAAA 2403:300:1364:0:8::1 AAAA 2403:3…
335 37.062647 2401:4900:50:9::7d5 2401:4900:8f56:7ec… DNS 264 Standard query response 0xf49b HTTPS gateway.fe2.apple-dns.net HTTPS NS ns-1396.awsdns-46.org NS ns…
382 37.754254 2401:4900:8f56:7ec… 2401:4900:50:9::7d5 DNS 134 Standard query 0x3a1e AAAA ap-sin-00001.s3.dualstack.ap-southeast-1.amazonaws.com
383 37.754363 2401:4900:8f56:7ec… 2401:4900:50:9::7d5 DNS 134 Standard query 0x02a5 A ap-sin-00001.s3.dualstack.ap-southeast-1.amazonaws.com
384 37.766096 2401:4900:50:9::7d5 2401:4900:8f56:7ec… DNS 495 Standard query response 0x3a1e AAAA ap-sin-00001.s3.dualstack.ap-southeast-1.amazonaws.com AAAA 240…
385 37.766102 2401:4900:50:9::7d5 2401:4900:8f56:7ec… DNS 399 Standard query response 0x02a5 A ap-sin-00001.s3.dualstack.ap-southeast-1.amazonaws.com A 52.219.16…
502 39.042771 2401:4900:8f56:7ec… 2401:4900:50:9::7d5 DNS 98 Standard query 0xc8c4 HTTPS prodregistryv2.org
503 39.042880 2401:4900:8f56:7ec… 2401:4900:50:9::7d5 DNS 98 Standard query 0xd8a6 AAAA prodregistryv2.org
504 39.042964 2401:4900:8f56:7ec… 2401:4900:50:9::7d5 DNS 98 Standard query 0x42aa A prodregistryv2.org
506 39.065112 2401:4900:50:9::7d5 2401:4900:8f56:7ec… DNS 180 Standard query response 0xc8c4 HTTPS prodregistryv2.org SOA ns-1528.awsdns-63.org
507 39.065114 2401:4900:50:9::7d5 2401:4900:8f56:7ec… DNS 180 Standard query response 0xd8a6 AAAA prodregistryv2.org SOA ns-1528.awsdns-63.org
509 39.095599 2401:4900:50:9::7d5 2401:4900:8f56:7ec… DNS 251 Standard query response 0x42aa A prodregistryv2.org A 34.128.128.0 NS ns-1528.awsdns-63.org NS ns-1…
567 39.570732 2401:4900:8f56:7ec… 2401:4900:50:9::7d5 DNS 113 Standard query 0xb9ba HTTPS sdmntprnortheu.oaiusercontent.com
568 39.570939 2401:4900:8f56:7ec… 2401:4900:50:9::7d5 DNS 113 Standard query 0x66af AAAA sdmntprnortheu.oaiusercontent.com
Frame 234: Packet, 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface
Ethernet II, Src: Apple_aa:1c:74 (c4:35:d9:aa:1c:74), Dst: ZyxelCommuni_ef:0a:ee (f4:4d:5c
Internet Protocol Version 6, Src: 2401:4900:8f56:7ecb:a854:7d07:a0c3:1e22, Dst: 2401:4900:
User Datagram Protocol, Src Port: 55854, Dst Port: 53
Domain Name System (query)
0000 f4 4d 5c ef 0a ee c4 35 d9 aa 1c 74 86 dd 60 00
0010 02 00 00 28 11 40 24 01 49 00 8f 56 7e cb a8 54
0020 7d 07 a0 c3 1e 22 24 01 49 00 00 50 00 09 00 00
0030 00 00 00 00 07 d5 da 2e 00 35 00 28 65 73 03 b1
0040 01 00 00 01 00 00 00 00 00 00 02 77 73 07 63 68
0050 61 74 67 70 74 03 63 6f 6d 00 00 41 00 01

Assignment - 5

Code -

```
from datetime import datetime

# Simulated honeypot logs
access_logs = [
    ("192.168.1.10", "SSH", "Unauthorized login attempt"),
    ("192.168.1.15", "HTTP", "Suspicious web request"),
    ("192.168.1.20", "SSH", "Multiple failed login attempts"),
    ("192.168.1.25", "FTP", "Unauthorized access attempt")
]

print("========== HONEYPOT MONITORING SYSTEM ==========")
print("Monitoring simulated suspicious connections...\n")

for ip, service, activity in access_logs:
    timestamp = datetime.now().strftime("%Y-%m-%d %H:%M:%S")

    print("Timestamp :", timestamp)
    print("Source IP :", ip)
    print("Service   :", service)
    print("Activity  :", activity)
    print("-" * 45)

print("\n========== HONEYPOT REPORT ==========")
```

```
print("Total suspicious activities recorded:", len(access_logs))
print("Services monitored: SSH, HTTP, FTP")
print("Alert: Suspicious access attempts detected.")
print("Monitoring completed.")
```

Output -

Output

```
========== HONEYPOT MONITORING SYSTEM ==========
Monitoring simulated suspicious connections...

Timestamp : 2026-10-13 15:16:50
Source IP : 192.168.1.10
Service   : SSH
Activity  : Unauthorized login attempt
-----------------------------------------
Timestamp : 2026-10-13 15:16:50
Source IP : 192.168.1.15
Service   : HTTP
Activity  : Suspicious web request
-----------------------------------------
Timestamp : 2026-10-13 15:16:50
Source IP : 192.168.1.20
Service   : SSH
Activity  : Multiple failed login attempts
-----------------------------------------
Timestamp : 2026-10-13 15:16:50
Source IP : 192.168.1.25
Service   : FTP
Activity  : Unauthorized access attempt
-----------------------------------------

========== HONEYPOT REPORT ==========
Total suspicious activities recorded: 4
Services monitored: SSH, HTTP, FTP
Alert: Suspicious access attempts detected.
Monitoring completed.
```

Assignment - 2

Code -

```
import random
import string

def generate_captcha():
    characters = string.ascii_uppercase + string.digits
    captcha = ''.join(random.choices(characters, k=6))
    return captcha

def verify_captcha(captcha):
    print("\nCAPTCHA Image Text:", captcha)
    user_input = input("Enter the CAPTCHA shown above: ")

    if user_input == captcha:
        print("CAPTCHA Verification Successful!")
    else:
        print("CAPTCHA Verification Failed!")

print("===== CAPTCHA GENERATION AND VERIFICATION =====")

captcha = generate_captcha()

verify_captcha(captcha)
```

Output -

Output

```
===== CAPTCHA GENERATION AND VERIFICATION =====

CAPTCHA Image Text: XBIJQW
Enter the CAPTCHA shown above: XBIJQW
CAPTCHA Verification Successful!
```

Output

```
===== CAPTCHA GENERATION AND VERIFICATION =====

CAPTCHA Image Text: Y87COA
Enter the CAPTCHA shown above: hhddff
CAPTCHA Verification Failed!
```